

Chapter 28. Home Mechanical Ventilation

Wolfram Windisch

Home Mechanical Ventilation: Introduction

Over the last three decades, home mechanical ventilation (HMV) has become a widely accepted treatment option for patients with chronic hypercapnic respiratory failure that arises from different etiologies such as chronic obstructive pulmonary disease (COPD), restrictive thoracic disorders, neuromuscular disorders, and obesity hypoventilation syndrome.^{1,2} There is increasing evidence that HMV is capable of improving symptoms, health-related quality of life (HRQL), and long-term survival in most of these patients,¹⁻³ although the impact of HMV on survival in patients with COPD is still a matter of debate.^{4,5} Thus, HMV should be considered in every patient presenting with symptomatic chronic hypercapnic respiratory failure.

In principle, HMV can be delivered by the application of long-term invasive mechanical ventilation, which requires the insertion of a tracheal tube following tracheostomy. Alternatively, noninvasive ventilation can be used to implement HMV via two possible routes. First, by application of negative pressure ventilation, which achieved international renown in the days when iron-lung ventilation was the preferred method for treating poliomyelitis; nowadays, this technique is seldom used.² Second, noninvasive positive-pressure ventilation (NPPV) can be delivered by connecting the natural airways of a patient and the artificial airways of the ventilator system by the use of face masks, which cover either the nose alone (nasal masks) or both the mouth and nose (oronasal masks).⁶ Mouthpiece ventilation is an additional option, particularly for neuromuscular patients.⁷

According to a large European epidemiologic study covering more than 21,000 HMV patients from sixteen different countries (Eurovent survey), the overall prevalence of HMV reportedly was 6.6 per 100,000 inhabitants.⁸ Substantial variation, however, has been identified among countries in terms of (a) prevalence, (b) the relative proportions of specific patient groups receiving HMV, and (c) HMV techniques. Nevertheless, the number of HMV patients is steadily increasing,⁹ and the Eurovent survey only refers to the time span of 2001 to 2002.⁸ Moreover, the survey was confined to selected HMV centers that were invited to participate, whereas HMV is increasingly being implemented by many hospitals that are not officially known as HMV centers. Thus, the prevalence of HMV varies substantially between different countries and is at present much higher, at least in some Western countries, than what was estimated by the Eurovent study. Of similar importance is the fact that the pattern of different conditions underlying chronic respiratory failure is changing over time, with patients suffering from COPD and obesity hypoventilation syndrome experiencing the largest increase in prevalence.⁹

The Eurovent study also identified 13% of the survey population as recipients of invasive ventilation, with the highest percentage comprising patients with neuromuscular disorders (24%). Most patients received NPPV and only 0.005%

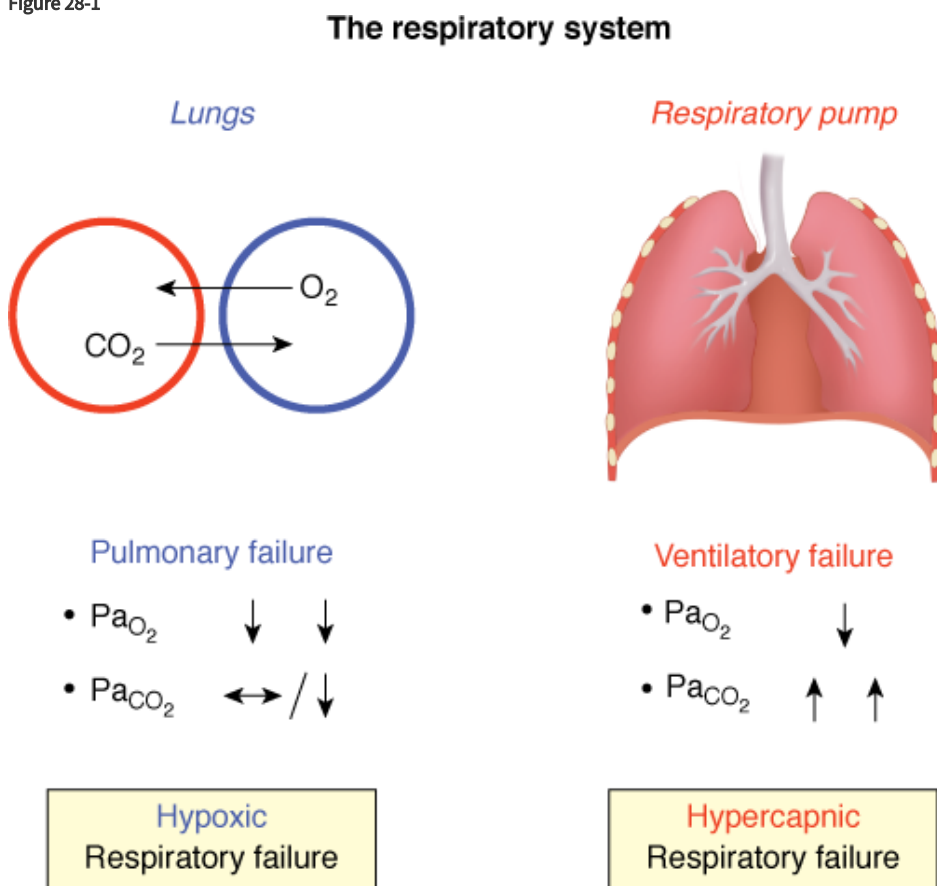
received other forms unlike positive-pressure ventilation. Thus, NPPV has become the predominant means of delivering HMV.

Rationale

The Respiratory System

The respiratory system consists of two independent parts, each of which can be selectively impaired by different pathologies (Fig. 28-1):¹⁰ (a) the lungs, which are responsible for gas exchange, and (b) the respiratory pump, which regulates mechanical movements to ventilate the lungs.

Figure 28-1



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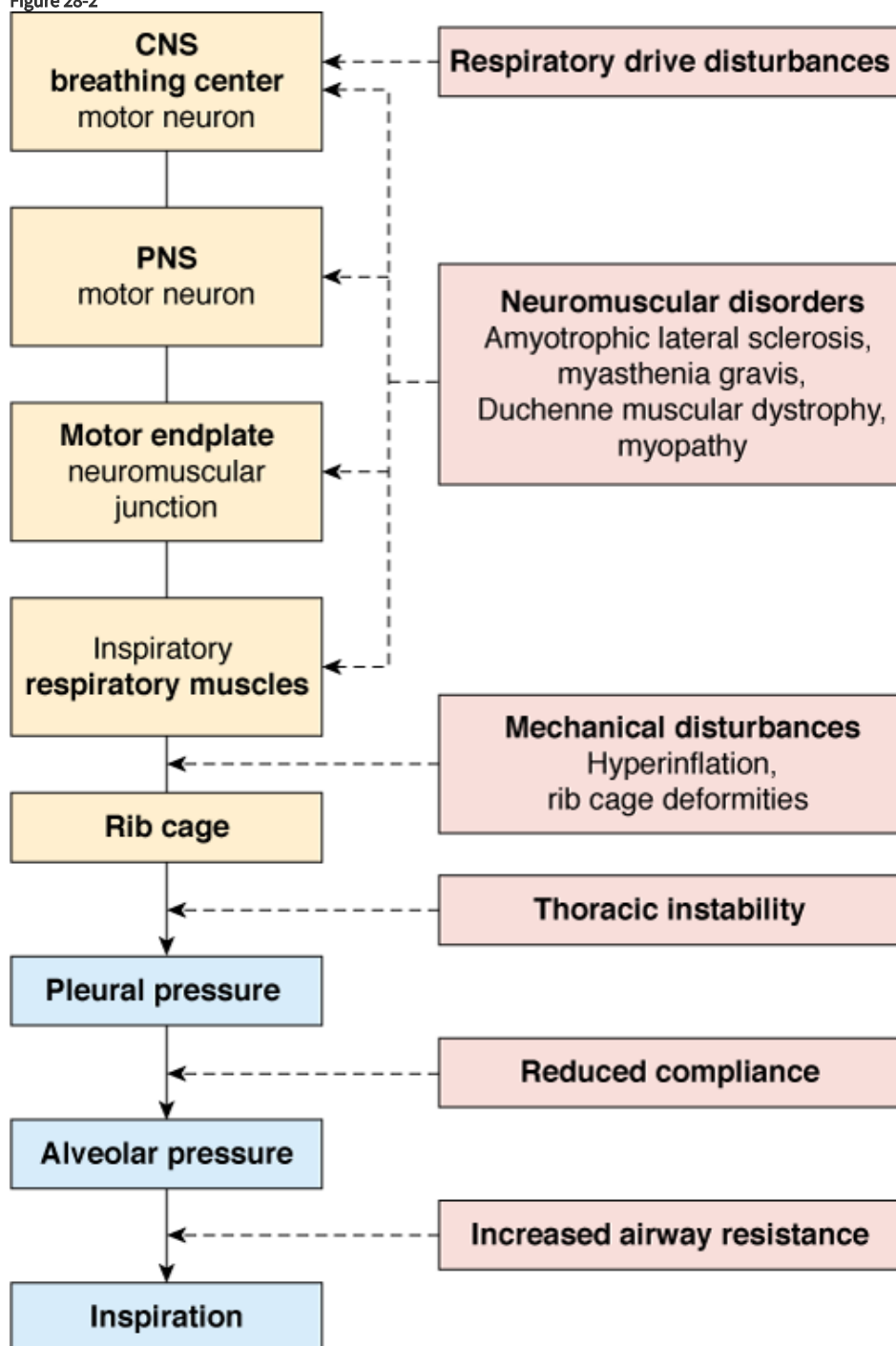
The two components of the respiratory system. In the lungs, **oxygen** (O_2) diffuses from the alveolus (*blue*) into the capillary (*red*), while carbon dioxide (CO_2) diffuses from the capillary into the alveolus. The respiratory muscles (*right panel*: diaphragm, intercostal muscles and accessory muscles) are essential for sufficient respiratory pump function. Pa_{O_2} , partial pressure of **oxygen**; Pa_{CO_2} , partial pressure of carbon dioxide.

Basically, pulmonary insufficiency leads to hypoxemic respiratory failure, indicating impaired gas exchange; here, **oxygen** is primarily affected because of its poorer diffusion capacities compared to carbon dioxide. In contrast, ventilatory insufficiency primarily leads to hypercapnia, indicating reduced alveolar ventilation, although hypoxemia also occurs as a result of hypoventilation. This is most often the result of increased load on the respiratory muscles, decreased respiratory muscle capacity, or both, although it can also result from decreased respiratory drive.^{10,11}

The Respiratory Pump

Treatment of chronic respiratory failure primarily depends on which part of the respiratory system is impaired. Chronic pulmonary failure is a well-justified basis for long-term [oxygen](#) treatment, with documented improvements in long-term survival in patients with COPD.^{12,13} In contrast, chronic failure of the respiratory pump coupled with reduced alveolar ventilation requires artificial augmentation of alveolar ventilation, which can only be achieved by long-term mechanical ventilation, that is, HMV. The indications for HMV are heterogeneous, in accordance with the complexity of the respiratory pump ([Fig. 28-2](#)).

Figure 28-2



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Primary components of the respiratory pump and the main conditions responsible for impairment, according to anatomical location. *CNS*, central nervous system; *PNS*, peripheral nervous system.

Mechanisms of Home Mechanical Ventilation

Several trials have established that in most patient groups, NPPV used for HMV is capable of improving physiologic variables, the most important of these being blood gases.^{1–4,9,14–20} Conflicting results in patients with COPD have been published; the effect of NPPV on blood gases was negligible in some trials,^{5,21–23} but a positive effect was clearly documented in other trials.^{3,4,24–27} NPPV improves blood gases not only during the time of application, but also during subsequent spontaneous breathing. This benefit has been attributed to an improved breathing pattern with increased tidal volume at an unchanged respiratory rate.¹⁸

Although the mechanisms by which intermittent NPPV improves subsequent spontaneous breathing remain unclear, three theories have been advanced: (a) respiratory muscle resting; (b) resetting of CO₂ sensitivity in the central breathing center; and (c) changes in pulmonary mechanics.^{1,2,20,28,29} The mechanisms are considered complementary rather than contradictory.^{1,2} It is, however, still unclear if these mechanisms are unique to all conditions leading to chronic ventilatory failure or are disease related. More recent research suggests that the principal mechanism in patients with restrictive thoracic disorders is an increased ventilatory response to CO₂.²⁰

Regarding the assessment of respiratory muscle strength, there is a lack of reliable methodology available. Volitional tests such as the widely used maximal inspiratory pressure method are highly dependent on the patient making a truly maximal effort.³⁰ In addition, normal values rely greatly on the study cohort and on several methodological aspects; this has resulted in contradictory regression equations for calculating normal values for maximal inspiratory pressure.³¹ Thus, this technique is not suitable for physiologic studies. In studies using nonvolitional gold-standard techniques, however, no changes in diaphragmatic muscle strength—as assessed by transdiaphragmatic twitch pressures following magnetic phrenic nerve stimulation—could be observed following initiation of HMV in restrictive thoracic disorders²⁰ and in COPD.³²

Another theory for improved spontaneous breathing stems from reports of improved lung function in patients with COPD in whom a substantial reduction in partial pressure of arterial carbon dioxide (PaCO₂) could be achieved.^{3,4,25,26} It has been speculated that the improvement results from a reduction in airway edema.^{25,33} Edema is a common finding in patients with COPD and might be related to *cor pulmonale* or possibly volume overload secondary to activation of sodium-retaining mechanisms. This, in turn, is thought to result from a reduction in effective circulating volume, secondary to a fall in total peripheral vascular resistance caused by hypercapnia-induced dilation of the precapillary sphincters.³⁴ A decrease in hypercapnia by means of HMV would then reverse dilation of the precapillary sphincters and thereby impact positively on the edema, which could improve respiratory mechanics in patients with airway edema.

Selection of Patients

Prerequisites for Home Mechanical Ventilation

There are three mandatory conditions for commencing HMV:^{1,2,35} (a) presence of an underlying condition or disease that potentially causes impairment of the respiratory pump and ultimately chronic ventilatory failure; (b) typical symptoms accompanying chronic ventilatory failure, despite optimal treatment; and (c) evidence of a chronically decompensated respiratory pump with reduced alveolar ventilation.

Underlying Condition and/or Disease

Before the commencement of HMV, the cause of the chronic respiratory pump failure needs to be identified and all appropriate treatment strategies instituted. It is necessary to bear in mind that more than one condition can be responsible for the development of chronic ventilatory failure in some patients. Chronic ventilatory failure from the combination of COPD and obesity is especially common. All diseases affecting the respiratory pump may warrant HMV treatment, and the most frequently reported conditions are listed in [Table 28-1](#).

Table 28-1: Conditions Most Commonly Managed by Home Mechanical Ventilation

Obstructive disorders
○ Chronic obstructive pulmonary disease
○ Cystic fibrosis
○ Bronchiectasis
○ Bronchiolitis obliterans
Restrictive thoracic disorders
○ Kyphoscoliosis
○ Posttuberculosis sequelae
○ Thoracoplasty
○ Restrictive pleural disease
○ Bechterew disease
○ Severe kyphosis
Neuromuscular disorders
○ Amyotrophic lateral sclerosis
○ Muscular dystrophy (Duchenne and others)
○ Critical illness polyneuropathy and myopathy
○ Paraplegia
○ Phrenic nerve paralysis
○ Postpoliomyelitis
○ Myopathy (congenital, metabolic)
○ Spinal muscular atrophy
○ Myasthenia gravis
○ Curschmann-Steinert disease
Obesity-hypoventilation syndrome
Central respiratory drive disturbances
○ Primary: Ondine's Curse
○ Secondary: brainstem alterations

Symptoms and Clinical Signs of Chronic Ventilatory Failure

No specific set of symptoms is reliably associated with chronic ventilatory failure because many occur in other conditions. In addition, symptoms ([Table 28-2](#)) depend on the underlying disease, its time course, comorbidities, and several other factors, such as medication and symptom perception. Some symptoms, however, are often falsely attributed to other conditions, leaving chronic ventilatory failure undiagnosed. Because one major goal of HMV is to alleviate symptoms, patients should be carefully screened for these symptoms both before and after the commencement of HMV ([Table 28-2](#)).

Table 28-2: Key Symptoms and Clinical Signs Related to Chronic Respiratory Failure

Symptoms and signs of the underlying disease
Symptoms and signs of the mechanical imbalance
Dyspnea
Tachypnea
Pathologic breathing pattern (depending on the underlying disease)
Symptoms and signs related to sleep disordered breathing
Tiredness
Daytime sleepiness
Nonrestful sleep
Poor concentration
Mental alterations
Symptoms and signs related to hypercapnia-induced vasodilation
Vasodilation in the eyes
Headache (particularly in morning)
Peripheral (airway) edema

Evidence of Chronically Reduced Alveolar Ventilation

The most important indicator of reduced alveolar ventilation is an elevated level of partial pressure of carbon dioxide (P_{CO_2}), which can be detected by arterial or capillary blood gas analysis, end-tidal recording, or transcutaneous measurement ([Table 28-3](#)).

Table 28-3: P_{CO_2} Monitoring

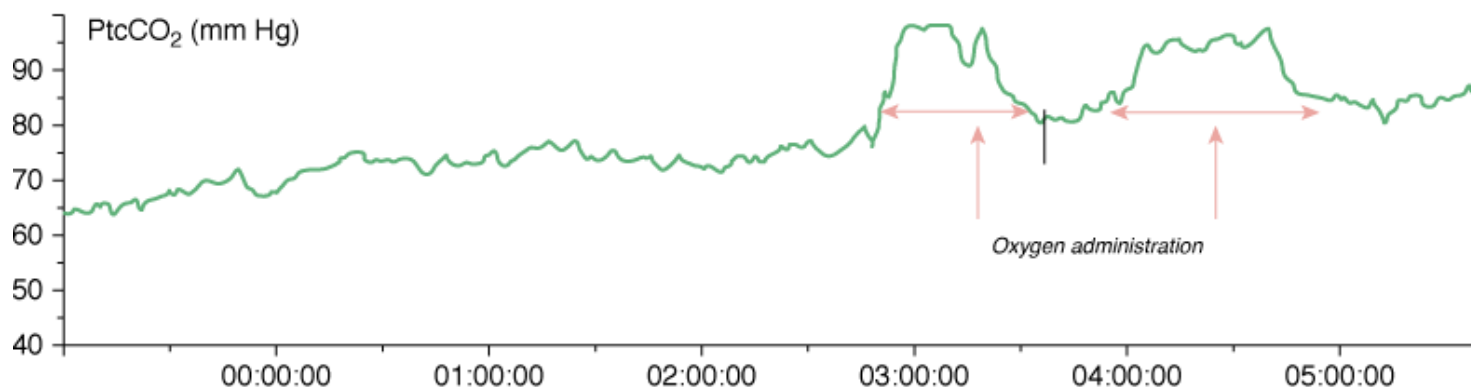
Method	Advantages	Disadvantages
Arterial (Pa_{CO_2})	• Reference standard • Other parameters coeval	• Invasive and painful • Selective
End-tidal (P_{ETCO_2})	• Continuous • Noninvasive	• Unreliable in case of leakage • Unreliable in $\dot{V}/\dot{Q}$ mismatch
Transcutaneous (P_{tcCO_2})	• Continuous • Noninvasive	• Drift of values over time • Problematic in skin abnormalities

Abbreviations: Pa_{CO_2} , arterial partial pressure of carbon dioxide; P_{ETCO_2} , end-tidal partial pressure of carbon dioxide; P_{tcCO_2} , transcutaneous partial pressure of carbon dioxide; $\dot{V}/\dot{Q}$, ratio of ventilation and perfusion.

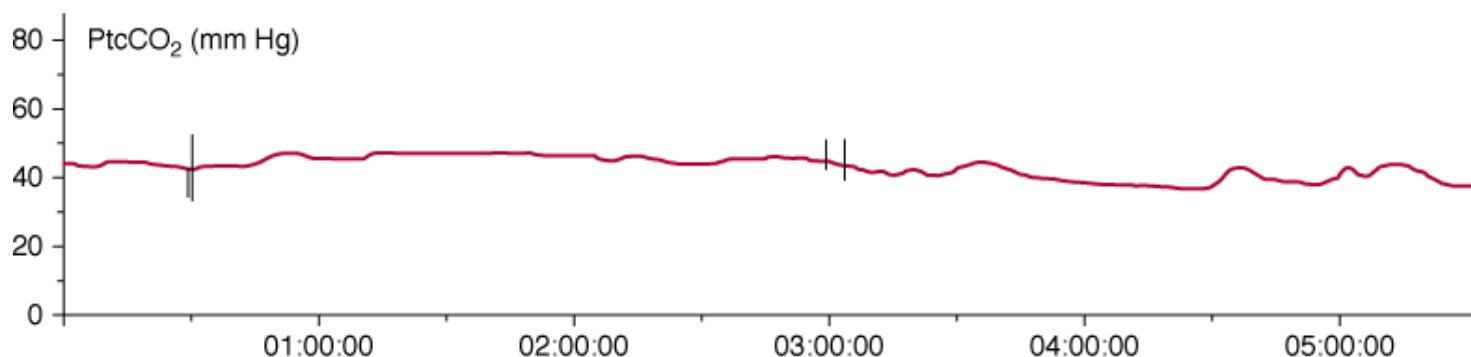
Although arterial measurement is regarded as the reference standard, transcutaneous measurements are increasingly recognized as being reliable. This has been attributed to technical refinements, which include the prevention and correction of measurement drifts.^{36,37} The advantage of transcutaneous measurements is that they enable noninvasive and continuous assessment. The technique is especially attractive during sleep, because sleep is not disturbed by an invasive procedure and it also captures any variations in alveolar ventilation (Fig. 28-3.)

Figure 28-3

A. Spontaneous breathing



B. Noninvasive ventilation



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Transcutaneous carbon dioxide tension (PtcCO₂) monitoring in a 49-year-old patient with chronic ventilatory failure secondary to amyotrophic lateral sclerosis. **A.** PtcCO₂ starts with values of 65 mm Hg during spontaneous breathing and progressively increases during sleep. Administration of oxygen produces an increase in PtcCO₂, likely caused by reduced alveolar ventilation. **B.** Home mechanical ventilation by means of noninvasive ventilation produces normalization of PtcCO₂.

Indications for Home Mechanical Ventilation

Hypercapnia

Initially, hypercapnia typically occurs during the night, and the patient may still be normocapnic during the day. Therefore, nocturnal measurements of P_{CO₂} are required to avoid overlooking the onset of chronic ventilatory failure. There are, however, no established cutoff values for nocturnal or daytime P_{CO₂} that would reliably indicate the need for HMV. Previous international recommendations have provided P_{CO₂} cutoff values for specific disease groups, but these recommendations are based on expert consensus rather than reliable research¹; moreover, this report only refers to NPPV, thus excluding invasive mechanical ventilation. The report¹ recommends long-term NPPV for restrictive patients when Pa_{CO₂} is equal to or higher than 45 mm Hg, while a Pa_{CO₂} equal to or higher than 55 mm Hg is recommended in patients with COPD. In cases where nocturnal desaturations occur, despite oxygen treatment, or in cases with recurrent hospitalizations in order to manage acute hypercapnic respiratory failure, a Pa_{CO₂} reading of

50 to 54 mm Hg warrants long-term NPPV.¹ Although detailed criteria are provided in the recently published German guidelines for HMV,³⁵ and a Canadian guideline is in preparation (personal communication), an international consensus on when to start HMV has not been reached. However, HMV needs to be carefully considered, not just in terms of PaCO₂ levels. Patients should be severely symptomatic with reduced quality of life, and should also be very likely to benefit from the therapy. This is particularly important for patients with COPD in whom clear improvements in long-term survival have yet not been scientifically established.^{5,22,23} Nevertheless, the use of HMV for preventive purposes in nonhypercapnic patients should be avoided.³⁸

For the commencement of HMV, chronic hypercapnic failure with normal pH (resulting from bicarbonate retention) should be established and distinguished from acute hypercapnic respiratory acidosis (Table 28-4).

Table 28-4: Blood-Gas Differences Depending on the Time Course of Ventilatory Failure

	Acute	Acute on Chronic	Chronic
pH	Lowered	Lowered	Normal
PaCO ₂	Increased	(Greatly) increased	Increased
HCO ₃	Normal	Increased	Increased

Abbreviations: HCO₃, bicarbonate; PaCO₂, arterial partial pressure of carbon dioxide.

Noninvasive versus Invasive Long-Term Mechanical Ventilation

In general, NPPV should be used in preference to invasive mechanical ventilation, which is used more often in neuromuscular patients following tracheostomy than in other disease groups.⁸ In particular, tracheostomy is indicated for the following conditions^{39–43}: lack of mask fitting; intolerance of, or claustrophobia with, mask ventilation; ineffective ventilation; impaired bulbar function with recurrent episodes of aspiration; insufficient noninvasive management of secretions requiring suction via a tracheal cannula; and failure to wean off invasive mechanical ventilation following intubation.

It is extremely important to comprehensively inform the patient and the family members about tracheostomy and its consequences. Patients’ self-determination is a top priority, and decision making should be based not only on objective circumstances but also on the subjective perspective of the patient.

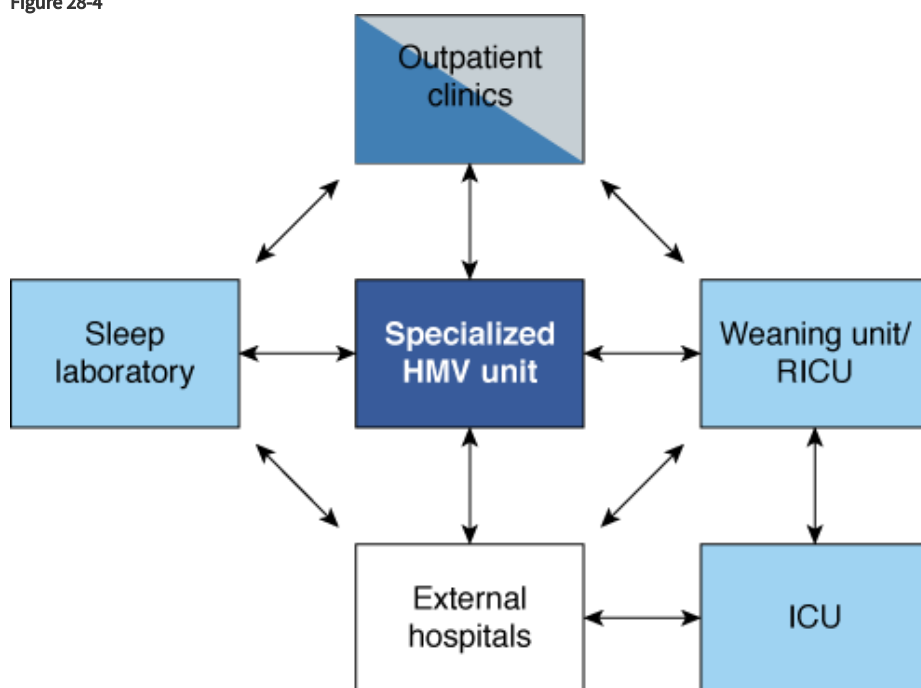
Organization

Specialized Center for Home Mechanical Ventilation

Patients receiving HMV should be managed in a specialized center that is familiar with the specific circumstances and problems occurring in such patients.³⁵ Here, high-quality, individually customized care is paramount, and it is essential that the treatment plan can be readily adapted to any necessary changes in the ventilator mode, settings, and interface, as well as in the duration, of mechanical ventilation. In all cases where the proposed treatment is subject to alteration, the patient, together with relatives and all associated therapists, must be included in the decision-making process, which must include close consultation and good organization between all participating individuals.

It is well recognized that the organizational structures vary considerably among countries and regions.⁸ Preferably, a specialized unit is set up exclusively for HMV and integrated into a network of specific units qualified for treating patients with mechanical ventilation (Fig. 28-4). Patients are usually admitted to the HMV unit from different sites within the same center, or from external sites (Fig. 28-4). Because many patients have sleep-disordered breathing, the center should provide easy access to full polysomnography, preferably within an in-house sleep laboratory. Many patients, particularly those with obesity hypoventilation syndrome, are primarily admitted to a sleep laboratory for diagnosis of sleep disturbances. In this case, sleep-related hypoventilation with deteriorating hypercapnia must not be overlooked, thus requiring close monitoring of nocturnal P_{CO_2} and the transfer to the HMV unit where appropriate.

Figure 28-4



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Specialized center for home mechanical ventilation. Patients are usually transferred to the HMV unit from other units within the specialized center, such as the sleep laboratory and weaning unit (*light blue*), or from external facilities (*white*) including outpatient clinics and external, nonspecialized hospitals. Return transfer should be facilitated at all times, according to patient's needs. *ICU*, intensive care unit; *RICU*, respiratory intensive care unit.

Especially confounding is the fact that HMV candidates are not always patients electively admitted to hospital: Some patients may also present with weaning failure consequent to ventilator management for acute respiratory failure (see Fig. 28-4). These patients are usually transferred from the intensive care unit to the HMV unit via a weaning unit

(“step-down”).⁴³ With the marked increase in the number of severely ill weaning-failure patients being admitted to a weaning unit,⁴⁴ close cooperation between the weaning and the HMV units is mandatory, because many weaning-failure patients are ultimately established on HMV.^{43,44}

Staffing and Training

Hospital staffing requirements depend on the type and severity of a patient’s underlying disorder. Staffing also relies on the local conditions of the center, such as experience, size, and country-specific organization of health care.⁸ Accordingly, great variation in staffing demands exists amongst different centers. Nevertheless, the global standards for staff include satisfactory training in mechanical ventilation techniques that covers both invasive and noninvasive ventilation. The team must be familiar with specific treatment modalities, such as airway clearance techniques, management of acute respiratory failure, and disease-specific issues.^{45,46} It is the responsibility of the center to provide sufficient staff training for all professionals involved, which may include: nurse, physician, physiotherapist, equipment provider, psychotherapist, respiratory therapist, and social worker.

In addition, a speech therapist, a nutritionist, or other therapist may be necessary at some point during the treatment of patients with HMV.

Transferring patients from hospital-based to home-based treatment is always a challenge.^{35,45–48} The challenge results from the necessity of replacing the in-hospital care team with an out-of-hospital care team, which may also include lay helpers, most importantly relatives. Training of all people caring for HMV patients in the home environment is essential.^{35,45,46} In addition, the specialized center and the provider should always be contactable for solving medical and technical problems, respectively.^{35,45,49}

Discharge from the Center and Transfer to Home Mechanical Ventilation

Both the underlying and secondary illnesses need to be in a stable condition before discharge.⁵⁰ If an optimal level of function and performance is not yet reached, (early) rehabilitative measures should be considered.^{35,45} In addition, the fulfilment of costs and provision of the necessary equipment, resources, and materials need to be ascertained in advance.³⁵ Individual requirements for patient discharge and subsequent setup of the home-ventilation station vary significantly from patient to patient. [Table 28-5](#) is a checklist of the minimal requirements.³⁵

Table 28-5: Checklist of Minimal Requirements for Patient Discharge and Subsequent Setup of a Home Ventilation Station

Full technical installation of the ventilator machinery and surveillance systems
Surveillance standards in terms of personnel (nurse attendance time)
Time schedule and content of nursing procedures
Type of ventilation interface and the corresponding cleaning and exchange intervals
Detailed description of ventilator mode and associated parameters
Duration of assisted ventilation and, if applicable, phases of spontaneous ventilation
Oxygen flow rates during assisted and spontaneous ventilation
Procedures for managing secretions
Application of inhaled medication
Planning for nutritional needs
Psychosocial care of the patient and, if applicable, the relatives
Additional therapeutic and educative measures
Additional resources (e.g., rollator, therapeutic bed, communication aids)

Based on Windisch et al.³⁵

Control Visits

Control visits include the collection of data on compliance, side effects, ventilation techniques, and daytime respiratory function, and also on nocturnal diagnostics, particularly nocturnal assessment of alveolar ventilation. Preferably, this should be performed in the specialized center where the patient has been initially treated. The first control examination after commencement of HMV should take place 4 to 8 weeks after discharge.^{35,47,51} Subsequent control visits typically occur once or twice a year and are carried out on an individual basis, depending on the type and progression of the underlying disease.

Specific Aspects of Noninvasive Ventilation Unique to Home Mechanical Ventilation

Impact of Ventilator Modes and Settings

Volume-Limited versus Pressure-Limited Noninvasive Positive-Pressure Ventilation

Volume-limited NPPV initially served as the primary mode for HMV, but pressure-limited NPPV has become the dominant mode,^{8,9} probably because constant peak inspiratory pressures (provided by pressure preset) are better tolerated than varying peak inspiratory pressures (provided by volume preset). In addition, ventilators providing pressure-limited ventilation are cheaper. According to the Eurovent study, volume-limited forms of mechanical ventilation were used in only 15% of patients with chronic lung diseases and in 28% of patients with restrictive thoracic disorders, although in a relatively higher number (42%) of patients with neuromuscular disorders.⁸

Physiologic studies show that leak-compensation capabilities, that is, increases in inspiratory flow in the event of air leakage, are only available for pressure-limited ventilator modes, and this holds true for both in vitro⁵² and in vivo⁵³ conditions. Early comparative trials revealed conflicting results, but interpretation of the findings is hindered by short observational periods, uncontrolled study design, and the lack of concomitant sleep studies.^{54–57} Two randomized, controlled, crossover trials, however, confirmed that the two modes were, on average, equally effective in improving alveolar ventilation during sleep, both in a mixed cohort that included patients with COPD¹⁷ and in a group of patients with restrictive thoracic disorders.⁵⁸ The first trial also indicated that pressure preset is better tolerated secondary to less gastrointestinal side effects,¹⁷ while the second trial suggested that switching from one mode to the other mode can provide individual benefits in terms of gas exchange.⁵⁸ Thus, there is no clear global advantage for one or the other mode, although changing the mode might be worthwhile for individual patients.

Target Volume

Most of the new-generation HMV ventilators provide the opportunity to set a so-called target volume. This expression describes a feature, in which the advantages of pressure preset (better tolerance because of less variation in peak inspiratory pressures) and volume preset (volume stability) are combined. Primarily, a pressure-limited mode is used, but instead of a fixed peak inspiratory pressure, a pressure range with minimal and maximal pressure is set at the ventilator. In accordance with the automatic calculations and adjustments of the ventilator, the pressure is thereby changed within a range so as to reach the preset target volume. Nomenclature and algorithms used for pressure adaption vary considerably among different types of ventilators.

Despite target volume being increasingly implemented in HMV ventilators, evidence supporting this new feature is sparse and long-term trials are nonexistent. The first randomized, controlled, crossover trial demonstrated that after 6 weeks of HMV using target volume settings, bi-level positive airway pressure ventilation was more effective, in comparison to settings without target volume, in improving gas exchange in patients with obesity hypoventilation syndrome; moreover, quality of life and polysomnographic sleep quality showed a comparable improvement following HMV.⁵⁹ Two further studies predominantly in patients with obesity-related chronic ventilatory failure revealed improvements in respiratory function through use of target volume settings^{60,61}; in one study, however, this occurred at the expense of objective and subjective sleep quality and comfort during ventilation.⁶¹ Finally, a randomized, controlled, crossover study on patients with COPD showed comparable results after 8 weeks of HMV.⁶²

High-Intensity Noninvasive Positive-Pressure Ventilation

High-intensity NPPV has been described as a new approach in patients with COPD (Table 28-6).^{4,18,25–27,36,53,63,64} In an attempt to decrease elevated PaCO_2 values, controlled ventilation with high inspiratory pressures (range: 20 to 40 cm H₂O) is implemented in the hospital setting and subsequently used for HMV. Several physiological and clinical variables are reportedly improved by this approach (Table 28-6). This technique contrasts with the conventional approach of using assisted ventilation with low inspiratory positive pressures (typically 10 to 16 cm H₂O), defined as low-intensity NPPV,^{5,21–23,65–68} which fails to effectively improve respiratory function. Direct comparisons following two randomized crossover trials clearly show the superiority of high- over low-intensity NPPV (Table 28-6).^{4,63} Consequently, high-intensity NPPV is the first choice for HMV in patients with COPD.

Table 28-6: High-Intensity Noninvasive Positive-Pressure Ventilation in Patients with Chronic Obstructive Pulmonary Disease

Definition	NPPV aimed at maximally reducing elevated PaCO_2 values with the major goal of achieving normocapnia or minimal PaCO_2 values using high ventilator settings, as maximally tolerated or necessary.
Procedure	- Mode: assisted or controlled, preferably pressure-limited, NPPV. - Use daytime NPPV first, with the primary aim of establishing tolerance, but also with control of blood gases and vital parameters. - Start with lowest backup respiratory rate and most sensitive trigger threshold (assisted NPPV); use low IPAP levels, typically ranging from 12 to 16 cm H_2O, and the lowest EPAP levels. - Then, carefully increase IPAP in a stepwise approach until maximal tolerance is reached, usually up to 30 (range: 20 to 40) cm H_2O. - Then, increase the respiratory rate just beyond the spontaneous rate to establish controlled ventilation. - Then, set EPAP so as to avoid dynamic hyperinflation according to subjective comfort (usually 3 and 6 cm H_2O), and similarly, set the I:E ratio to 1:2 or lower. - Once daytime tolerance is acceptable, apply nocturnal NPPV. - Adjustments to ventilator settings are made according to subjective tolerance and nocturnal monitoring of blood gases.
Pitfalls and practical advice	- Tolerance of higher IPAP levels can last from minutes to several days; individual adjustment is inevitable. - In cases of coexisting upper airway obstruction, higher EPAP levels are required. On the other hand, higher EPAP reduces the effective IPAP (which is IPAP minus EPAP); thus, avoid high EPAP levels if not required. - For controlled NPPV (final aim), respiratory rates are typically set to 1 breath/min higher than during spontaneous breathing; thus, avoid excessively high respiratory rates, although try to establish controlled ventilation. - Try out several masks: for nocturnal NPPV, use oronasal masks because of potentially substantial leakage; for daytime NPPV, a nasal mask is often better tolerated. Again, individual adjustment is mandatory. - Several days in hospital are usually necessary to establish high-intensity NPPV. - Use humidification in cases of dry mucous membrane. - Leakage is unavoidable, but should be kept as low as possible. - Gastrointestinal side effects can be managed by medication, positioning, and adjustment (reduction) of ventilator settings.
Physiologic effects	- Improvement in blood gases during NPPV.^{4,18,26,27} - Improvement in blood gases during subsequent spontaneous breathing.^{3,4,18,25–27} - Improvement in breathing pattern during spontaneous breathing.¹⁸

	- Improvement in lung function.^{3,4,25,26} - Improvement in global inspiratory muscle strength.¹⁸ - Increases in hematocrit in anaemic patients.²⁵ - Reduces hematocrit in patients with polyglobulia.²⁵ - Superior to low-intensity NPPV using assisted ventilation with low IPAP regarding the improvement of blood gases.^{4,63}
Clinical effects	- Improvement in health-related quality of life.^{3,4,18} - Improvement in dyspnea while spontaneous breathing during walking.⁴ - Improvement in dyspnea and walking distance during NPPV-aided walking compared to unaided walking.⁶⁴ - Acceptable sleep quality.^{17,63} - Superior to low-intensity NPPV using assisted ventilation with low IPAP regarding adherence to therapy.⁴
Important unknowns	Improvement of long-term survival?

Abbreviations: EPAP, expiratory positive airway pressure; I:E, inspiratory-to-expiratory timing; IPAP, inspiratory positive airway pressure; NPPV, noninvasive positive-pressure ventilation; Pa_{CO_2} , arterial partial pressure of carbon dioxide.

Quality Control of Ventilator Settings at Home

For HMV to be successful, it is essential that patients use the prescribed equipment as well as the ventilator modes and settings that were established in the hospital. Changing the system or the settings can reduce the quality of ventilation and result in either hypoventilation or hyperventilation. For this reason, a quality assessment of ventilator performance at home has been proposed.⁵¹ A recent survey including 326 centers visited by more than 20,000 patients in sixteen European countries revealed significant deficits in quality control; only 56% of centers assessed whether the patients and/or caregivers correctly cleaned or maintained the ventilator.⁵¹ Considerable intercountry and intracountry differences were observed, and the size of the center was a major determinant of several quality-control aspects.⁵¹ Considerable differences were found between several prescribed versus actually implemented ventilator variables, and important alarm functions, such as power off, disconnection, and obstruction, were sometimes defective.⁴⁸

Different interfaces used for HMV have varying physical characteristics. This is particularly true for the resistance component, which is known to be flow-dependent, and applies to interfaces used for both noninvasive⁶⁹ and invasive mechanical ventilation.⁷⁰ Therefore, changing the interface in the home environment without control

measures can produce alterations in the quality of HMV. Indeed, the type of interface has been shown to affect NPPV outcome even more than the ventilator mode.⁷¹ Thus, changes to the ventilator system, the settings or interface should be done with adequate control in cooperation with the HMV-prescribing center.

Side Effects

Most side effects of HMV arise from the mismatch between the artificial airway and the natural human airway, although complications may also be caused by the ventilator itself. Side effects are highly dependent on whether invasive or noninvasive HMV is used (Table 28-7). Although side effects are common,^{2,3,17} NPPV is safe. Nevertheless, side effects can impair HRQL and necessitate the discontinuation of ventilator treatment. Every effort should be made to minimize side effects. For example, humidification should be provided in cases of dry mouth or throat, and facial soreness or erythema or ulceration, resulting from mask pressure, can be ameliorated by minimizing strap tension or switching to an alternative interface. Despite frequently reported side effects, NPPV can improve symptoms and HRQL in many patients.³

Table 28-7: The Most Relevant Side Effects of Home Mechanical Ventilation

Noninvasive Home Mechanical Ventilation	Invasive Home Mechanical Ventilation
<i>Mask-related:</i> • Facial and/or nasal soreness • Facial and/or nasal erythema or rash • Facial and/or nasal ulceration • Gum pain	<i>Canula-related:</i> • Tracheal injury and/or bleeding/pain • Tracheal stenosis and/or tracheomalacia • Tracheal granulation tissue • Tracheal stenosis and/or fistula • Dislocation and/or obstruction • Difficulties in speaking and/or coughing • Infections
<i>Airflow and air pressure or leakage related:</i> • Dry nose and/or throat and/or bronchial tree • Nasal congestion • Eye irritation • Ear and/or sinus pain • Epistaxis • Gastric insufflation and/or distension • Belching and/or flatulence • Abdominal pain • Nausea/vomiting	<i>Airflow and air pressure related:</i> • Barotrauma • Volutrauma
<i>Sleep problems:</i> • Not falling asleep • Sleep disruption	<i>Sleep problems:</i> • Not falling asleep • Sleep disruption

The frequency and dimension of side effects depends on several factors including the ventilator modes, settings and interfaces. For example, volume-limited NPPV produces more gastrointestinal side effects because of higher and more variable peak inspiratory pressures as compared with pressure-limited NPPV.¹⁷ Side effects of NPPV tend to diminish once patients become more familiar with HMV.³

Outcome

Long-Term Survival

There is overwhelming evidence that following the commencement of HMV, long-term survival is markedly improved in patients with restrictive thoracic diseases and neuromuscular disorders, particularly those that are slow-progressing,^{19,72} although the effect of HMV on outcome in patients with COPD is still a matter of debate.^{4,5,21–23} Evidence for improved survival in restrictive and neuromuscular patients is derived mostly from uncontrolled trials,

but the data are sufficiently robust enough to support the conclusion that survival is increased in these patients. Randomized controlled trials in these patient groups presumably will never be conducted because of ethical concerns.⁷²

Detailed information for six diseases typically treated by HMV follows.

Early Kyphoscoliosis

In landmark observational trials of the 1990s, 1-year survival rates reached 90% in patients with kyphoscoliosis following the commencement of NPPV treatment.^{14,15} The 5-year survival rate was almost 80%,^{9,15} and the 9-year survival rate reported by the Swedish registry reached 55%.⁷³ Survival was better in kyphoscoliosis patients receiving HMV compared to patients receiving long-term oxygen treatment, even though the HMV patients were deemed to be more ill.

Posttuberculosis Sequelae

Five-year survival rates in patients with posttuberculosis sequelae were higher than 90% in one study,¹⁵ although survival was considerably lower (50%) in two other trials.^{9,74} This discrepancy is likely related to differences in baseline ages of patients: mean age was 61 years in the first trial, but 72 years⁷⁴ and 75 years⁹ in the other two trials. The Swedish registry revealed that approximately 21% of patients had a 9-year survival rate when HMV was started at a baseline age of 72 years.⁷⁴ Uncontrolled data of the Swedish registry reveals that survival in posttuberculosis sequelae patients who received HMV was higher than in those who received oxygen treatment.⁷⁴

Duchenne Muscular Dystrophy

The mean age at death in patients with Duchenne muscular dystrophy is approximately 19 years if long-term mechanical ventilation is not instituted, and most die from ventilatory failure.⁷⁵ Although invasive mechanical ventilation,⁷⁵ mouthpiece ventilation,⁷⁶ and negative pressure ventilation⁷⁷ have shown beneficial effects on long-term outcome, NPPV is considered to be the treatment of choice.⁷⁸ Given that mean survival without HMV is less than 10 months once chronic hypercapnic respiratory failure has occurred,⁷⁹ the reported 5-year survival rate of 73% in patients with Duchenne muscular dystrophy who received long-term NPPV is phenomenal.⁸⁰ Thus, HMV must not be withheld in these patients. These positive results encouraged researchers to apply prophylactic long-term NPPV in nonhypercapnic patients. Unfortunately, NPPV did not ameliorate the respiratory handicap and even reduced survival.³⁸ Consequently, prophylactic HMV must be avoided. If NPPV is used, additional techniques for clearance of secretions are often required; if invasive ventilation is used, tracheal suction can easily be performed via a canula. The use of NPPV in addition to technically assisted coughing prolongs survival while significantly decreasing the pulmonary morbidity and hospitalization associated with conventional invasive technical ventilation.⁸¹

Amyotrophic Lateral Sclerosis

In contrast to Duchenne muscular dystrophy, amyotrophic lateral sclerosis tends to progress more rapidly. In addition, bulbar involvement may prevent the patient from receiving NPPV, although tolerance may be feasible in patients with mild-to-moderate bulbar involvement.^{78,82} In severe and rapidly progressive neuromuscular disorders, as with amyotrophic lateral sclerosis, fears exist that use of HMV in advanced disease might simply protract death rather than prolong good quality of life.¹⁹ Nevertheless, early, uncontrolled studies suggest that survival can be improved by the addition of NPPV,^{82–85} and one randomized controlled trial demonstrated that NPPV improves survival and HRQL in patients with amyotrophic lateral sclerosis, although the survival benefit was restricted to patients without severe bulbar dysfunction.⁸⁶

Obesity-Hypoventilation Syndrome

Uncontrolled data suggest that prognosis of untreated obesity-hypoventilation syndrome is poor,^{87–90} with a mortality rate of 23% after 18 months in one trial.⁹⁰ This contrasts with more recent long-term data on 126 patients showing 1-, 2-, and 5-year survival rates of 97%, 92%, and 70%, respectively, in patients treated with NPPV. A reduction in nocturnal PaCO_2 by 23% or more was associated with improved survival rates in this trial.⁹¹ Large randomized controlled, long-term trials do not exist, and several issues still need to be addressed, including the need to define clear treatment goals.⁸⁷ It is also unclear which type of ventilator support should be used in which patient. Recent data has convincingly shown that continuous positive airway pressure and bi-level ventilator support were equally effective in improving daytime hypercapnia in patients without severe nocturnal hypoxemia,⁹² although differences in the effects on long-term survival still need to be established.

Chronic Obstructive Pulmonary Disease

In contrast to restrictive and neuromuscular patients, HMV has not been convincingly shown to improve survival in COPD. The approximate 5-year survival rate is 43%, which is much lower than in restrictive patients (excluding those with rapidly progressive diseases).¹⁴ Two randomized controlled trials showed no survival benefit.^{22,23} An Australian randomized controlled trial observed an improvement in survival, but this applied to the adjusted, but not to the unadjusted model; this minor effect occurred at the cost of worsening HRQL.⁵ Low-intensity NPPV was used in most randomized controlled trials, and also failed to effectively improve respiratory function.^{5,21–23} One could argue that improved survival cannot really be expected if the specific modality fails to improve physiology, that is, if NPPV is relatively ineffective. In contrast, high-intensity NPPV has indeed been shown to improve several physiological and clinical variables^{3,4,18,25–27,63,64} and 5-year survival (58%) was higher than that previously reported.¹⁸ This promising approach needs to be tested for survival benefits in randomized controlled trials.

Health-Related Quality of Life

How Can Health-Related Quality of Life Be Measured in Patients with Home Mechanical Ventilation?

Evaluation of HRQL provides an important means of evaluating the human and financial costs and benefits of modern medical treatment modalities.^{93–95} HRQL evaluation is particularly important for patients with chronic and

noncurable disorders, and from a patient's point of view, the effect of HMV on HRQL may be more important than its effect on long-term survival.

Questionnaires are the most frequently used tool for HRQL assessment, and it is crucial that essential psychometric properties—the most important being objectivity, reliability, validity, sensitivity, and responsiveness—are established before they are used as an HRQL assessment tool in clinical trials.⁹³ Although generic questionnaires are not tailored to a particular disease and can thus be applied comparably in different diseases, they lack sufficient responsiveness and sensitivity to specific treatment interventions; thus these instruments only qualify for comparative, cross-sectional HRQL assessments. The opposite is true for disease- or condition-specific instruments, which are primarily designed to longitudinally assess changes in HRQL, but restricted to the particular condition for which they are developed.⁹³ Table 28-8 lists the instruments most frequently used in HMV cohorts.

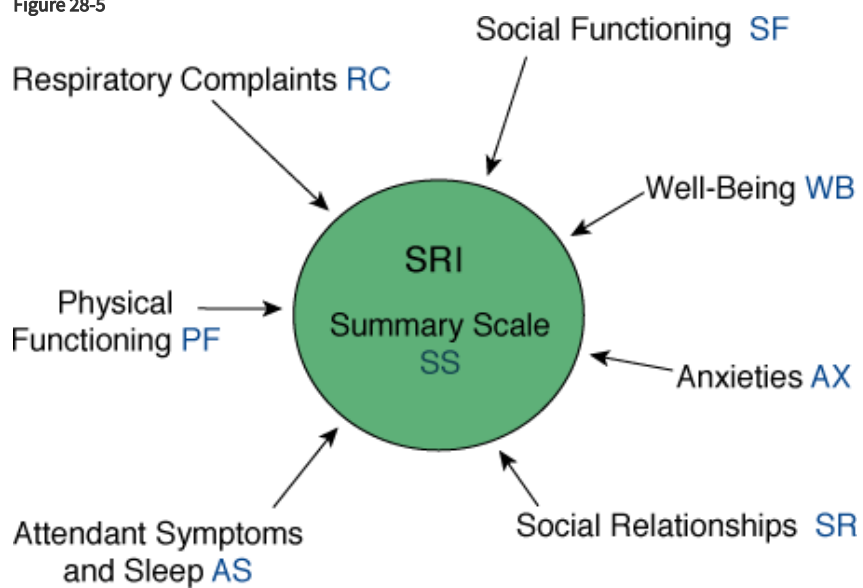
Table 28-8: The Most Frequently Used Instruments for Assessment of Health-Related Quality of Life in Patients Receiving Home Mechanical Ventilation

Instrument	First Author	Subscales (n)	Items (n)
Generic instruments			
Sickness Impact Profile (SIP) ^{96,97}	Bergner M	12	136
Nottingham Health Profile (NHP) ⁹⁸	Hunt SM	6	38
Hospital Anxiety and Depression Scale (HAD) ⁹⁹	Zigmond AS	2	14
MOS 36-Item Short-Form Health Survey (SF-36) ^{100–102}	Ware JE	8	36
COPD-specific instruments			
Chronic Respiratory Disease Questionnaire (CRQ) ¹⁰³	Gyatt GH	4	20
St. George's Respiratory Questionnaire (SGRQ) ¹⁰⁴	Jones PW	3	76
Instruments specific for chronic respiratory failure			
Maugeri Foundation Respiratory Failure item set (MRF-28) ¹⁰⁵	Carone M	3	28
Severe Respiratory Insufficiency questionnaire (SRI) ^{106,107}	Windisch W	7	49

The Severe Respiratory Insufficiency Questionnaire is a multidimensional health measure (Fig. 28-5), developed exclusively for patients receiving HMV.^{106,107} It is valid for a broad variety of diseases, including restrictive thoracic and neuromuscular disorders, and obesity hypoventilation syndrome,¹⁰⁶ and additional research supports the

questionnaire's validity for patients with COPD.¹⁰⁷ Intercultural adaption is currently in progress, with ten professional translations in the process of validation or consideration (Table 28-9.)

Figure 28-5



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation*, 3rd Edition: www.accessanesthesiology.com

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The multidimensional Severe Respiratory Insufficiency (SRI) Questionnaire with seven subscales.

Table 28-9: Intercultural Adaption of the Severe Respiratory Insufficiency Questionnaire

Language	Status	Literature (Refs.)
German	Original version validated	106 , 107
Spanish	Translated version validated	108 , 109
English	Translated version validated	110
Norwegian	Translated version validated	111
Dutch	Translation finished	112
French	Translation finished	113
Swedish	Translation finished	
Greek	Translation finished	
Japanese	Translation finished	
Hebrew	Translation in progress	

Impact of the Underlying Disease on Health-Related Quality of Life

The first study to systematically analyze HRQL in HMV patients readily established that the underlying disease is a main determinant of HRQL, with patients suffering from posttuberculosis sequelae experiencing worse HRQL than other restrictive patients, particularly those with kyphoscoliosis.^{[114](#)} This study indicated that patients who are stable while receiving HMV have only minor impairments in psychosocial components of HRQL, similar to that of patients with other chronic diseases. The underlying disease has also been shown to be a determinant of HRQL in patients with chronic ventilatory failure who are not receiving HMV.^{[115](#)} More recent research demonstrated that patients with COPD and posttuberculosis sequelae have more HRQL impairments than patients with other restrictive thoracic or neuromuscular disorders,^{[106](#),[116](#),[117](#)} and that mental health can be well preserved, despite severe physical handicaps, particularly in patients with Duchenne muscular dystrophy.^{[116](#)} HRQL as assessed by the Severe Respiratory Insufficiency Questionnaire was an overall predictor of long-term survival in patients without COPD receiving HMV.^{[117](#)}

Impact of Home Mechanical Ventilation on Health-Related Quality of Life

Numerous studies have investigated the influence of HMV on HRQL. Many of these trials used generic HRQL measures because specific instruments were not widely available at the time. Interpretation of these trials is limited because

generic instruments do not cover the specific issues relevant to patients with chronic respiratory failure and are postulated to be less sensitive to any changes occurring.⁹³

Compared to the effect on survival, there is little doubt that HMV substantially improves both HRQL (as assessed by the Severe Respiratory Insufficiency Questionnaire)^{3,59} and sleep quality^{16,17,59} in patients with restrictive diseases. In contrast, longitudinal studies, mostly using COPD-specific HRQL measures, have only recorded minor and inconsistent improvements in HRQL following HMV in patients with COPD.^{118–120} One randomized controlled trial reported that application of the COPD-specific St. George's Respiratory Questionnaire revealed no differences in HRQL when HMV plus long-term oxygen therapy was compared to long-term oxygen therapy alone; in contrast, application of the more condition-specific Mageri Foundation Respiratory Failure item set showed that HMV improved HRQL.²³ Thus, even COPD-specific instruments may fail to reliably show HRQL changes in patients with severe COPD. This observation is probably attributable to the fact that these instruments were developed for patients with milder COPD.¹⁰⁵ Nevertheless, the Severe Respiratory Insufficiency Questionnaire revealed substantially improvements in HRQL in patients with COPD following commencement of HMV.^{3,4,18} These improvements were similar to those seen in restrictive patients. Therefore, robust data support the conclusion that HMV can improve HRQL in several different disease groups including patients with COPD.

Important Unknowns and Future Considerations

Despite HMV being fully accepted as a treatment option for patients with chronic respiratory failure, some important questions remain open.

First, new features and modes are being developed by the manufacturers of home ventilators. These are aimed at not only improving the tolerance and quality of ventilation, but also at facilitating the application of home ventilators. Scientific evidence, however, for these developments is lacking. A good example is target volume. Its rationale is clear and target volume has been implemented in various ventilators, although robust evidence to support its use is not available. New features with new names and abbreviations inevitably overwhelm the user. From a technical point of view, HMV has been extremely successful for decades and although technical refinements are warranted, the challenge for future research is to differentiate between the innovations that help patients and those that cause confusion without any benefit.

Second, a pressing issue to determine circumstances in which HMV can prolong life in patients with COPD. Here, the new approach of high-intensity NPPV is promising and needs testing with regards to its potential for improving survival. It is not just different ventilator strategies that need to be investigated; clearly specified criteria for starting HMV also need to be established. Recent research has demonstrated that institution of HMV following an acute exacerbation complicated by respiratory acidosis and the need for acute mechanical ventilation was capable of preventing recurrent clinical deterioration.^{121,122} Whether further acute readmissions to hospital can be prevented by HMV application and whether this impacts survival is a question that needs to be elucidated in the future.

Third, HMV undoubtedly improves HRQL. Nevertheless, the clinical status of patients eventually deteriorates, ultimately leading to death, and there is no conclusive information about end-of-life issues and decision making in the terminal phase of seriously ill HMV patients. There is only one study, which indicates that many HMV patients during their last 3 months of life frequently had respiratory symptoms and received inadequate medication.¹²³ The

family burden was high and more than 50% of patients were eventually hospitalized and many died in the hospital or even in the intensive care unit. Research should not only focus on physiologic, epidemiologic, technical, structural, financial, and outcome aspects, but also study how patients with HMV eventually die and how HRQL in the terminal phase can be improved so as to enhance the quality of dying.¹²⁴

Summary and Conclusion

HMV has become a standard treatment for patients suffering from chronic ventilatory failure, with the most frequently treated diseases being COPD, restrictive thoracic disorders, neuromuscular disorders, and obesity-hypoventilation syndrome. NPPV using face masks has become the modality of choice, although some patients, particularly those with neuromuscular disorders and difficulties in clearing their secretions, still use invasive mechanical ventilation via tracheostoma, which enables tracheal suctioning.

HMV should be started when a disease or a particular circumstance is causing impairment of the respiratory pump, and there is evidence of chronic ventilatory failure together with symptoms related to chronic hypoventilation, especially dyspnea and sleep-disordered breathing, headache, or edema. Ventilation should be commenced in specialized centers with suitable experience. Staff training and close collaboration between the ventilation center and the people caring for the patient in the home environment are essential for successful treatment. HMV has been shown to improve symptoms, quality of life, and long-term survival in most of these patients, but its full impact on long-term survival in patients with COPD still needs to be established. Finally, understanding how patients receiving HMV die and how the quality of dying can be improved needs to be elucidated in the future.

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